

Alec Portelli

Active U.S. Secret Clearance

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Work Experience

The MITRE Corporation

Robot Software Engineer II • April 2023 - Present

Robot Software Engineer I • July 2021 - April 2023

Leading development for a simulation testbed to train a UAV with AI. Helping create an autonomy framework that uses ROS 2 and Isaac Sim. Helping write control and mapping algorithms. Conducted over ten demonstrations to NVIDIA, the Navy, Microsoft, and MITRE executives which secured over a year's funding.

Building a scenario designer using Unity to help engineers simulate electronic warfare environments. Developing software architecture and writing physics models for UAVs. Released software to over 30 users. Time needed to build and simulate scenarios decreased by 300%.

Researched AI security by training a model rocket to land with a camera in Isaac Sim and then hacking the algorithm. Helped write pipelines to connect algorithm data to security evaluation toolkits. Presented findings to MITRE chief engineers.

Built a electronic warfare visualization tool in Unity to help analyze aviation and radio simulations. Wrote plane and UAV simulators to accelerate hardware in-the-loop testing. Led a team of three engineers and deployed software to over 20 users.

UserWise LLC

Engineering Intern • October 2020 - March 2021

Wrote MATLAB simulations to evaluate product safety for medical device startup. Designed UIs and manuals using human factors best practices.

Selected Masters Program Projects

UAV Systems And Controls

Wrote a 6DoF simulation model of a UAV using Simulink and MATLAB. Built a UAV from scratch using an Arduino. Designed and implemented a control system and wrote navigation algorithms.

Intro To Reinforcement Learning

Created a model UAV quadcopter in Unity and trained it to navigate an unknown environment. Used trained model to simulate package delivery systems amongst multiple UAVs.

Engineering Skills

Robot Simulation

Machine Learning

3D Modeling

Simulation and Modeling

Control Design

Mapping / SLAM

Software Architecture

UI / UX Design

Arduino

Certifications

AWS Cloud Practitioner

Tools

MATLAB

Simulink

Isaac Sim

ROS 2

Gazebo

Unity

PyBullet

Git

Autodesk Suite

AWS

Languages

Rust

C#

Python

JavaScript

SQLite

C++

Education

Johns Hopkins

University

Masters: Robotics

Tufts University

Bachelors: Engineering

Double major in
human-computer interaction
and biomedical engineering